

# Emran Masoumifeshani, PhD

Sr. AI Data Scientist | AI/ML Engineer

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## Profile

PhD-trained scientist and AI/ML engineer with hands-on experience building Generative AI applications, agentic systems (LangChain, LangGraph), LLM evaluation frameworks, and model lifecycle governance, complemented by a strong background in computational modeling, statistical analysis, and data engineering (GCP, Databricks, SQL/NoSQL). Brings 7+ years of professional experience across industrial R&D, scientific computing, and applied AI/data engineering.

Proven track record designing scalable computational methodologies and delivering reproducible, data-driven solutions for large-scale scientific challenges – including high-performance computing and cross-functional collaboration – resulting in publications in leading journals such as *Angewandte Chemie*, *Journal of Chemical Theory and Computation*, and *Journal of Computational Chemistry*. Skilled in explainable AI (SHAP), bias and fairness assessment, and regulatory compliance (GDPR): well suited to AI engineering, Model Risk Management, and applied AI roles within highly regulated industries such as financial services.

## Core Skills

- **AI/ML & Model Risk Management (MRM):** Generative AI & LLMs (LangChain, LangGraph), Independent Model Validation (IMV), LLM Evaluation (LangSmith, DeepEval), LLMOps, Model Lifecycle Governance, Explainability (SHAP), Bias & Fairness, GDPR / Regulatory Compliance
- **Agentic AI Systems:** Design and evaluation of agent-based architectures and autonomous decision frameworks
- **Programming:** Python, C++, R, JavaScript
- **Data & Cloud:** GCP BigQuery, Databricks, SQL, NoSQL (MongoDB), Spark, PySpark, ETL, Data Modeling, Vector Databases, Azure, AWS
- **ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, Model Versioning (MLflow)
- **Analytics & Experimentation:** A/B Testing, Causal Analysis, Statistical Modeling
- **Data Platforms & BI:** Power BI, Tableau, Looker Studio
- **Engineering & MLOps:** FastAPI, Airflow, Docker, CI/CD, Git, API Testing (Postman), JIRA, Confluence

## Professional Experience

**Founder | Sr. AI Data Scientist | AI/ML Engineer**

Feb 2025 – Present

*Royal Dywany, Warsaw*

- Led design, development and evaluation of agentic AI architectures using LangChain and LangGraph to automate decision workflows and business processes, cutting manual turnaround time.
- Built and validated LLM evaluation pipelines (LangSmith, DeepEval) covering accuracy, robustness, explainability (SHAP), and bias/fairness assessment prior to deployment across two business-facing AI applications.
- Directed the company's data platform strategy across GCP BigQuery and Databricks, including ETL pipeline design, data modeling, and integration of SQL/NoSQL (MongoDB) and vector databases for retrieval-augmented generation (RAG).
- Implemented core MLOps practices including model versioning, Docker-based deployment, API development and CI/CD workflows.
- Implemented model lifecycle governance and GDPR-aligned data compliance processes, and introduced BI reporting (Power BI, Looker Studio) for stakeholder decision-making.
- Managed a cross-functional engineering team of 5 and delivery workflow using Git, JIRA and Confluence, overseeing product strategy alongside technical execution.

## Sr. Research Assistant & Assistant Professor

*The Institute of Organic Chemistry, Polish Academy of Sciences (IOC PAS), Warsaw*

Sep 2023 – May 2025

- Conducted advanced computational research involving large-scale scientific datasets, predictive modelling, statistical analysis, and validation of computational models.
- Designed and implemented Python-based computational workflows for automated data processing, feature extraction, simulation pipelines, and reproducible scientific experiments.
- Applied rigorous model evaluation and benchmarking methodologies to assess accuracy, robustness, uncertainty, and reliability of computational models.
- Developed data-driven approaches for analysing complex molecular systems, integrating numerical simulations, statistical modelling, and scientific machine learning concepts.
- Managed computational experiments through reproducible workflows, documentation practices, and systematic validation procedures aligned with model lifecycle principles.
- Collaborated with interdisciplinary research teams, translating complex computational outputs into actionable insights and contributing to peer-reviewed publications.

## Doctoral Researcher

*University of Warsaw, Warsaw*

Dec 2018 – May 2023

- Conducted doctoral research in computational modelling, algorithm development, statistical analysis, and data-driven investigation of complex molecular and material systems.
- Designed and developed the **AROFRAG** fragmentation methodology for systematic decomposition of aromatic molecular systems, improving computational efficiency and accuracy of interaction modelling.
- Developed Python-based automated pipelines for data generation, preprocessing, computational experiments, statistical analysis, model benchmarking, and result interpretation.
- Performed extensive model validation, performance evaluation, and benchmarking using Density Functional Theory (DFT) and *ab initio* methods, applying quantitative metrics to assess model accuracy, robustness, uncertainty, and generalisation capability.
- Applied statistical modelling, uncertainty analysis, quantitative data analysis, and hypothesis-driven evaluation techniques to investigate intermolecular interactions in supramolecular and nanomaterial systems.
- Served as a **Teaching Assistant for Computational Chemistry and Quantum Chemistry courses**, supporting students in computational methods, quantitative problem-solving, statistical analysis, and interpretation of computational results.
- Analysed high-dimensional simulation and experimental datasets using statistical and computational methods, identifying patterns, quantifying relationships, and translating complex data into scientifically meaningful conclusions.
- Designed reproducible computational workflows using version-controlled environments and structured research processes to ensure transparency, reliability, and reproducibility of scientific analyses.
- Communicated complex computational and statistical results through peer-reviewed scientific publications and international conference presentations.

## Chemistry Teacher

*British International School, Warsaw*

Nov 2019 – Jan 2020

- Delivered chemistry lessons in English within an international curriculum; planned and adapted engaging lessons for a diverse student body.
- Assessed student progress and maintained effective classroom management in a short-term teaching role.

## Research & Development Engineer

*Nirou Trans Company, Shiraz, Iran*

Nov 2013 – Nov 2018

- Evaluated and qualified raw material applications for polymer and composite systems, assessing composition and properties against technical/safety data sheets in coordination with procurement, engineering and quality teams.
- Conducted chemical resistance, mechanical (tensile, elongation, flexural, hardness) and rheological/viscoelastic characterisation of polymer and elastomer systems.
- Executed accelerated ageing and thermal endurance studies under defined stress conditions, tracking material property changes over time.
- Developed and validated new test methods, drafting test programmes, acceptance criteria and SOPs for laboratory use.

## Education

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### Doctor of Philosophy (Ph.D.) in Chemical Sciences

2018–2023

University of Warsaw, *Warsaw, Poland*

- Dissertation: Developed a novel computational methodology for systematic molecular fragmentation of branched molecules and polycyclic aromatic hydrocarbons, enabling scalable quantum chemical modeling of large molecular systems.

### Master of Science (M.Sc.) in Physical Chemistry

2011–2013

University of Tehran, *Tehran, Iran*

- GPA: **18.66/20** **Ranked 1st in Class**
- Thesis: Determination of Potential Energy Curves for OH, CH<sup>+</sup>, and NeH<sup>+</sup> Molecules Using Spectroscopic Data

### Bachelor of Science (B.Sc.) in Chemistry

2007–2011

University of Tehran, *Tehran, Iran*

- GPA: **17.41/20** **Ranked 1st in Class**

## Publications

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- A. Pareek, M. Morawiak, **E. Masoumifeshani**, P. Gawel, “Photostable  $\pi$ -Expanded Furans via Base-Mediated Quinone Oxygen-Annulation,” *The Journal of Organic Chemistry*, 2026, **91**, 2837–2843. doi:10.1021/acs.joc.5c02906
- M. Y. Mehboob, **E. Masoumifeshani**, Z. Badri, C. Foroutan-Nejad, “Design of Carbon–Carbon Ylides,” *Angewandte Chemie International Edition*, 2025, **64**, e202507446. doi:10.1002/anie.202507446
- K. Bartkowski, **E. Masoumifeshani**, M. Kotowska, U. Klimczak, B. Furman, C. Foroutan-Nejad, M. Lindner, “One-Pot Transition-Metal-Free Synthesis of  $\pi$ -Extended Bipolar Polyaromatic Hydrocarbons,” *Angewandte Chemie International Edition*, 2025, **64**, e202423282. doi:10.1002/anie.202423282
- **E. Masoumifeshani**, T. Korona, “Intermolecular Interaction Energies with AROFRAG—A Systematic Approach for Fragmentation of Aromatic Molecules,” *Journal of Computational Chemistry*, 2024, **45**(29), 2446–2458. doi:10.1002/jcc.27429
- **E. Masoumifeshani**, T. Korona, “AROFRAG—A Systematic Approach for Fragmentation of Aromatic Molecules,” *Journal of Chemical Theory and Computation*, 2024, **20**, 1078–1087. doi:10.1021/acs.jctc.3c00875
- T. Korona, J. Jankowska, **E. Masoumifeshani**, “Dicarbon Defect in Hexagonal Boron Nitride Monolayer—A Theoretical Study,” *Canadian Journal of Chemistry*, 2023. doi:10.1139/cjc-2022-0291
- **E. Masoumifeshani**, M. Chojecki, D. Rutkowska-Zbik, T. Korona, “Association Complexes of Calix[6]arenes with Amino Acids Explained by Energy Partitioning Methods,” *Molecules*, 2022, **27**, 7938. doi:10.3390/molecules27227938
- **E. Masoumifeshani**, M. Chojecki, T. Korona, “Electronic Correlation Contribution to the Intermolecular Interaction Energy from Symmetrized Systematic Molecular Fragmentation Model,” *Computational and Theoretical Chemistry*, 2022, **1211**, 113684. doi:10.1016/j.comptc.2022.113684
- **E. Masoumifeshani**, T. Korona, “Symmetrized Systematic Molecular Fragmentation Model and Its Application for Molecular Properties,” *Computational and Theoretical Chemistry*, 2021, **1202**, 113303. doi:10.1016/j.comptc.2021.113303

## Certifications

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- Building Generative AI-Powered Applications with Python – IBM
- Generative AI Language Modeling with Transformers – IBM
- Gen AI Foundational Models for NLP & Language Understanding –IBM
- Generative AI and LLMs: Architecture and Data Preparation –IBM
- The Structured Query Language (SQL) – University of Colorado Boulder
- Foundations: Data, Data, Everywhere – Google